



Name: _____ Student No.: _____ Course/Section: _____ Date: _____



LEARNING OBJECTIVES

After completing this topic, students should be able to:

- 1 Define population and sample.
- 2 Distinguish between parameter and statistic.
- 3 Identify common statistical symbols.
- 4 Explain the basic idea of sampling distribution.
- 5 Differentiate population measures from sample measures.



A. CONCEPT CHECK

Write the letter of the term that best matches each description.

- _____ 1. The entire set of individuals or items of interest in a study.
- _____ 2. A subset of the population selected for a study.
- _____ 3. A numerical value that describes a population.
- _____ 4. A numerical value computed from a sample.
- _____ 5. The variability of statistics from all possible samples of the same size.

Choices

- A. Population
- B. Sample
- C. Parameter
- D. Statistic
- E. Sampling Distribution



B. APPLICATIONS

For each scenario below, identify whether the given value is a population measure (parameter) or a sample measure (statistic). Write your answer on the blank.

No.	Scenario	Parameter	Statistic
1	The mean height of all college students in the university is 168 cm.	☒ (Population mean, μ)	☐
2	The average monthly allowance of 50 randomly selected students is ₱4,250.	☐	☐
3	The proportion of all households in a city that have internet access is 0.72.	☐	☐
4	A random sample of 100 voters shows that 60% prefer Candidate A.	☐	☐
5	The standard deviation of the weights of all apples produced in a farm is 15 grams.	☐	☐



C. STATISTICAL SYMBOLS

Identify the meaning of each symbol below. Choose your answer from the box.

- Population mean
- Population standard deviation
- Population size
- Sample mean
- Sample standard deviation
- Sample size
- Population proportion
- Sample proportion

Symbol	Meaning	Symbol	Meaning
1. μ	_____	5. s	_____
2. σ	_____	6. n	_____
3. N	_____	7. p	_____
4. $\bar{X}$	_____	8. $\hat{p}$	_____



D. SHORT ANSWER

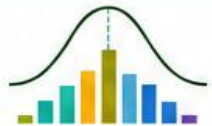
Answer the following questions briefly but clearly.

1. What is a sampling distribution? Give an example.

2. Why is the sampling distribution of a statistic important in inferential statistics?

3. Differentiate between a parameter and a statistic. Give an example for each.





A. POPULATION, SAMPLE, PARAMETER, OR STATISTIC?

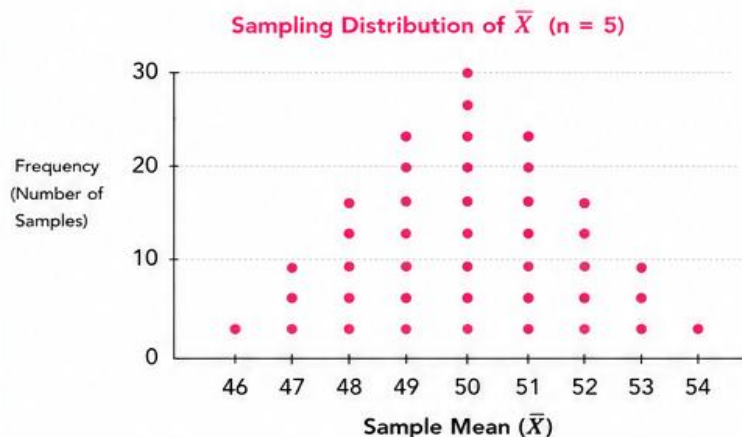
For each situation below, identify the population, the sample, the parameter of interest, and the statistic that will be computed.

No.	Situation	Population (Who or what?)	Sample (Units selected)	Parameter (What we want)	Statistic (What we compute)
1	A university wants to estimate the average monthly allowance of all full-time students. A random sample of 80 students is selected and their average monthly allowance is computed.	All full-time students in the university	The 80 students randomly selected	The population mean monthly allowance (μ)	The sample mean monthly allowance ($\bar{X}$)
2	A company wants to estimate the proportion of defective light bulbs in a shipment of 10,000 bulbs. A random sample of 200 bulbs is tested.				
3	A health agency wants to estimate the mean systolic blood pressure of all adults in a city. A random sample of 120 adults is selected and their average systolic blood pressure is computed.				



B. INTERPRETATION: SAMPLING DISTRIBUTION

The dot plot below shows the sampling distribution of the sample mean $\bar{X}$ for all possible simple random samples of size $n = 5$ drawn from a population with mean $\mu = 50$.



- What is the center (mean) of the sampling distribution of $\bar{X}$?

- What is the shape of the sampling distribution?

- Will the mean of the sampling distribution of $\bar{X}$ be equal to the population mean μ ? Why or why not?

- If the sample size were increased to $n = 25$, how would the spread (variability) of the sampling distribution of $\bar{X}$ likely change? Explain.



C. APPLICATION (BASIC)

Solve or answer the following.

- A population has mean $\mu = 120$ and standard deviation $\sigma = 24$.
A random sample of size $n = 36$ is taken.
Find the mean and standard deviation (standard error) of the sampling distribution of the sample mean $\bar{X}$.

Solution:

Mean of $\bar{X}$: _____

Standard deviation (SE) of $\bar{X}$: _____





A. HIGHER ORDER THINKING (HOTS)

- 1 A population has mean $\mu = 90$ and standard deviation $\sigma = 18$. A random sample of size $n = 49$ is drawn.

a. What is the mean of the sampling distribution of $\bar{X}$?

b. What is the standard deviation (standard error) of $\bar{X}$?

c. If the sample size is increased to $n = 196$, what happens to the standard error of $\bar{X}$? Explain why.

- 2 A city's daily rainfall amounts (in mm) are highly skewed to the right.

a. If all possible simple random samples of size $n = 30$ are taken from this population, what can we say about the shape of the sampling distribution of the sample mean $\bar{X}$? Why?

b. Will the sampling distribution of $\bar{X}$ be more spread out or less spread out if the sample size is increased to $n = 120$? Justify your answer.



B. REFLECTION

- 1 What did you learn from this topic?

- 2 What concept was most challenging for you? Why?

